

Phase 2: Methodological Limitations, Statistical Weaknesses & Strategic Roadmap

Analytical Audit Context & Objectives

- **What is being evaluated:** Methodological assumptions, statistical power constraints, biological abstractions, and computational limits across Phase 2 (Univariate Associations, Youden Cutoffs, and Genomic Interaction Modelling).
- **Why we are doing it:** Critical assessment of standalone biomarker thresholds and interaction models is necessary to establish realistic boundaries for clinical translation and prevent over-interpretation of univariate metrics.
- **What question it answers:** What statistical, biological, and computational constraints limit the standalone utility of Phase 2 cutoffs, and how can future pipeline iterations resolve them?

1. Executive Summary & Audit Overview

Phase 2 evaluates biomarker discriminative power across $N_{ICI} = 326$ immunotherapy-treated melanoma patients using non-parametric Mann-Whitney U testing, Cohen's d effect sizes, Youden's J optimal decision thresholds, and 21 driver mutation $\times$ immune signature logistic regression interaction terms.

While Phase 2 establishes key empirical benchmarks—such as `B_cells` operating as the top standalone predictor ($AUC = 0.632$, cutoff $= 0.430$) and `BRAF` oncogenic signalling significantly dampening T-cell inflammation ($TIS \times BRAF \beta = -0.65, p = 0.040$)—several statistical, biological, and data-structure constraints limit direct clinical application. This audit details these constraints in the current pipeline state and outlines actionable, prioritised technical solutions.

Summary of Core Phase 2 Analytical Limitations

- **Step-Function Information Loss:** Discretising continuous biomarker distributions into binary cutoffs via Youden's J discards continuous biological variance and creates artificial boundary misclassifications.
- **Uncorrected Multiple Testing:** Evaluating 21 interaction pairs simultaneously without False Discovery Rate (FDR) control risks Type I errors among secondary interaction signals.
- **Bulk RNA-seq Spatial Blindness:** Bulk transcriptomic `B_cells` scoring cannot determine spatial organisation into Tertiary Lymphoid Structures (TLS).
- **Subgroup Sample Size Imbalance:** Statistical power is heavily skewed toward `mut_BRAF` ($N = 128$), leaving smaller subgroups (`mut_NF1` $N = 40$, `mut_NRAS` $N = 82$) underpowered for interaction detection.

2. Code-Quality & Architectural Considerations

Evaluating the current Phase 2 software architecture (`02_feature_analysis.py` and `src/feature_analysis.py`) highlights several design considerations for scalable pipeline development:

1. Tight Coupling of Execution Script and Output Paths:

- `02_feature_analysis.py` hardcodes output CSV export paths (`univariate_feature_associations.csv`, `youden_cutoffs.csv`, `genomic_immune_interactions.csv`) directly to `data/processed/q5/`.
- **Impact:** Hinders programmatic parameter sweeps or multi-dataset benchmarking without overwriting baseline processed artifacts.

2. In-Memory Transformation Overhead:

- Univariate Mann-Whitney U testing and Youden ROC iterations recreate subset DataFrames in memory per feature loop rather than vectorising operations across feature blocks.
- **Impact:** Increases execution overhead when expanding the feature panel beyond the core baseline markers.

3. Fallback Error Handling in Interaction Fits:

- When statsmodels `Logit` fails to converge on sparse feature subsets, the exception handler falls back to returning $\beta_{interaction} = 0.0$ and $p = 1.0$.
- **Impact:** While preventing pipeline crashes, silent fallback masks convergence issues caused by severe feature collinearity or extreme class imbalance in smaller driver subgroups.

3. Statistical Weaknesses & Methodological Constraints

Methodological & Statistical Constraints

The following statistical properties constrain the diagnostic precision and generalisability of Phase 2 cutoffs:

1. Information Loss from Step-Function Threshold Discretisation

- **Current Approach:** Youden's J statistic calculates single rigid numerical cutoffs (such as `B_cells` ≥ 0.430 or `TIS` ≥ 0.191) to classify continuous score distributions into binary High vs Low risk categories.
- **Statistical Criticism:** Converting continuous biological variables into step-function binary categories discards quantitative variance. A patient scoring 0.429 is classified as Low risk alongside a patient scoring -1.500 , despite 0.429 lying immediately adjacent

to the 0.430 cutoff threshold.

- **Impact:** Step-functions misclassify borderline patients and fail to capture the smooth, continuous gradient of immune activation in real-world tumours.

2. Uncorrected Multiple Testing Across Interaction Matrices

- **Current Approach:** Logistic regression interaction models test 21 driver mutation $\times$ microenvironment signature pairs independently, reporting raw unadjusted p -values.
- **Statistical Criticism:** Testing 21 hypotheses simultaneously without Benjamini-Hochberg False Discovery Rate (FDR) correction inflates the family-wise Type I error rate. While $\text{TIS} \times \text{BRAF}$ reaches strict significance ($p = 0.040$), borderline interactions ($\text{B_cells} \times \text{BRAF}$ $p = 0.073$, $\text{CD8_T_cells} \times \text{BRAF}$ $p = 0.074$, $\text{IFN_gamma} \times \text{BRAF}$ $p = 0.087$) require strict multi-testing control to confirm true biological validity.
- **Impact:** Risks over-interpreting exploratory secondary interaction terms in unpowered subgroups.

3. Modest Standalone Diagnostic Accuracy (AUC $\approx 0.58 - 0.63$)

- **Current Approach:** Evaluating continuous biomarkers as standalone univariate response predictors.
- **Statistical Criticism:** Even the top individual feature (B_cells) achieves an AUC of only 0.632 (Sensitivity = 53.7%, Specificity = 77.0%), while core T-cell signatures (TIS AUC = 0.585, CYT AUC = 0.583, CD8_T_cells AUC = 0.584) yield modest separation.
- **Impact:** Demonstrates mathematically that single-biomarker triage is insufficient for clinical selection, providing the empirical rationale for multi-dimensional clustering in Phase 3.

4. Biological Assumptions & Clinical Translation Limits

⚠ Biological Abstractions & Clinical Translation Boundaries

Biological simplifications inherent in bulk sequencing and retrospective trial data limit direct clinical translation:

1. Bulk RNA-seq Blindness to Tertiary Lymphoid Structures (TLS)

- **Biological Constraint:** Transcriptomic cell deconvolution identifies B_cells as the top standalone predictor ($d = 0.373$, AUC = 0.632). However, bulk RNA sequencing measures total tissue RNA abundance and cannot determine whether B cells are organized into functional **Tertiary Lymphoid Structures (TLS)** with germinal centres versus unorganised, non-functional naive infiltrates.
- **Clinical Implication:** Mature TLS containing co-localised B cells (CD20^+), T cells (CD8A^+), and follicular dendritic cells (CD21^+) drive anti-PD-1 response. Two tumours with identical bulk B_cells scores may exhibit opposite clinical outcomes based on spatial TLS maturation.

2. MAPK Signalling Abstraction & BRAF Interaction Mechanism

- **Biological Constraint:** The interaction model confirms that BRAF mutation dampens T-cell predictive value ($\beta = -0.65$, $p = 0.040$). However, linear interaction terms abstract the underlying biochemistry—constitutive BRAF V600E signalling suppresses antigen presentation (B2M , HLA-A/B/C) and secretes immunosuppressive cytokines (VEGF , IL-6 , IL-10).
- **Clinical Implication:** A static linear interaction coefficient misses non-linear threshold dynamics where high MAPK flux completely suppresses immune recognition below a critical T-cell infiltration level.

3. Unmeasured Prior Systemic Therapy Confounding

- **Biological Constraint:** Pre-treatment biopsies from clinical trials ([Liu 2019](#), [Riaz 2017](#), [Hugo 2016](#)) are evaluated without detailed records of prior targeted kinase inhibitor exposure or washout duration.
- **Clinical Implication:** Prior BRAF/MEK inhibition (Dabrafenib/Trametinib) transiently increases T-cell infiltration and HLA expression. Unmeasured prior therapy introduces unobserved confounding into baseline biomarker evaluations.

5. Computational & Subgroup Data Constraints

⚠ Data Structure & Subgroup Imbalances

Subgroup sample size disparities restrict statistical detection power across driver subtypes:

1. Driver Subgroup Sample Size Imbalance

- **Data Constraint:** ICI cohort sample sizes vary substantially across driver mutation subgroups:
 - mut_BRAF : $N = 128$ (39.3% of cohort)
 - mut_NRAS : $N = 82$ (25.2% of cohort)
 - mut_NF1 : $N = 40$ (12.3% of cohort)
- **Computational Impact:** High sample size in mut_BRAF provides strong power to detect interaction terms ($\text{TIS} \times \text{BRAF}$ $\beta = -0.65$, $p = 0.040$). Conversely, mut_NF1 displays the largest positive interaction effect size with macrophage polarisation (M1_M2_Ratio $\beta = +0.94$), but yields $p = 0.654$ due to statistical underpowering ($N = 40$).

2. Missing Co-stimulatory Mappings in IMPRES Calculation

- **Data Constraint:** IMPRES evaluates 15 pairwise boolean comparisons between co-stimulatory and co-inhibitory genes. Several key co-stimulatory genes (CD28 , CD86 , CD80 , CD40 , CD200) are missing or unmapped in legacy cohort matrices.
- **Computational Impact:** Reduces IMPRES calculation to a truncated pair subset, systematically underestimating checkpoint regulation in specific historical trial arms.

6. Actionable Prioritised Technical Fixes & Strategic Roadmap

To resolve these statistical and biological constraints in future iterations, we propose five concrete technical enhancements:

Priority	Constraint Category	Proposed Technical Solution	Implementation Tool / Library	Expected Analytical Benefit
P1	Threshold Discretisation	Replace binary Youden step-functions with Generalized Additive Models (GAMs) and restricted cubic splines.	pyGAM / statsmodels.gam	Preserves continuous risk gradients; eliminates artificial boundary misclassification.
P2	Multiple Testing	Implement Benjamini-Hochberg FDR control ($q < 0.05$) and Bayesian interaction shrinkage.	statsmodels.stats.multitest	Controls false discovery rates across multi-permutation interaction matrices.
P3	Spatial B-Cell Function	Combine transcriptomic deconvolution with spatial TLS image quantification via multiplex immunofluorescence.	QuPath / 10x Xenium / StarDist	Distinguishes functional Germinal Centre TLS from unorganised B-cell infiltrates.
P4	Subgroup Imbalance	Apply balanced bootstrap resampling and power-weighted interaction regression adjustments.	Scikit-learn Resampling / Custom PyTorch	Equalises statistical detection sensitivity across BRAF , NRAS , and NF1 cohorts.
P5	Prior Therapy Confounding	Annotate clinical metadata with prior targeted therapy duration and washout intervals.	Pandas Clinical Pipeline	Eliminates unobserved confounding from prior BRAF/MEK inhibitor exposure.

7. Key Takeaways

📌 **Key Takeaways: Phase 2 Limitations & Strategic Future Roadmap**

- **Step-Functions Discretise Variance:** Binary Youden cutoffs provide useful clinical triage numbers (e.g. $B_cells \geq 0.430$) but discard continuous biological information, necessitating GAM spline modelling in future work.
- **Univariate Tests Are Modest:** Single-biomarker predictive power tops out at $AUC = 0.632$ (B_cells), proving mathematically that single-gene tests fail in complex tumours and validating Phase 3 multidimensional clustering.
- **Genomic Interactions Require Power:** BRAF oncogenic signalling significantly dampens T-cell inflammation ($\beta = -0.65, p = 0.040$), but smaller driver subgroups ($NF1, N = 40$) require power-weighted resampling to confirm high effect-size synergies ($\beta = +0.94$).
- **Spatial Resolution Is Essential:** Bulk transcriptomics cannot confirm TLS spatial maturation; future clinical translation requires integrating spatial histology image quantification.